Stat231: Blog Project

Project Description

Summary

In collaboration with your group, you will embark on a data science project with the overarching goal of advancing your data science skills and experiencing how empowering data science can be. You will create a publicly available blog that explores a topic of interest to your group. In addition to increasing your skills and your confidence in data science, I hope this project helps you recognize the role of data science in tackling important issues, and providing you an important, meaningful lens in understanding your world and the current world we're in, nationally and internationally.

This project is deliberately open-ended to allow you to explore your creativity and interests. There are only three main rules that must be followed:

1. **Your project must be centered around messy data.** Preferably, you will work with large, complex and/or messy data. Alternatively, you may work with a dataset that isn't very messy, but is challenging to obtain (e.g., a challenging web scraping task).
2. **You must incorporate (at least) one of the methods introduced in the second half of the semester: spatial data, text analysis, network science, and/or unsupervised learning.** Although we only spend a week or less introducing each of these topics, there are entire courses dedicated to each one! This project provides you an opportunity to take a deeper dive into one or more of these exploratory analyses.
3. **Your project must tell us something thought-provoking and meaningful.** On one extreme are *data art* projects like the Dear Data project or Memo Akten's *Forms*, which may involve little to no statistical analysis. On the other extreme are data mining projects like the KDD Cup annual data mining competition, which may involve little to no visualization. Your project can be anywhere on this spectrum, and expectations may be different depending on where you are on the scale. Your project should go beyond downloading a single data source and summarizing it with with some perfunctory visualization. Make sure there is depth to the statistical exploration (Rule 2) and meaning.

Learning Objectives

The learning objectives of the project is to demonstrate your ability to:

1. Identify a set of questions that can be addressed with data available to you
2. Wrangle a large, messy dataset (including gathering, reshaping, and cleaning the data) into a format necessary to answer the question(s) at hand
3. Create effective data visualizations
4. Effectively communicate results in written and oral form
5. Effectively collaborate with your peers and identify the value in teamwork
6. Demonstrate awareness of ethical considerations related to data acquisition, management, and communication

Timeline

All deliverables must be submitted by 9:00 PM (unless otherwise indicated) on the dates provided at the appropriate submission location. Adherence to these deadlines is required to help us keep on pace for the semester.

Activity	Date	Submission
Initial Proposal	Thursday, Oct. 15	New GitHub Issue (Blog plan)
Revised Proposal	Tuesday, Oct. 20	New comment on GH Issue (Blog plan)
Wrangled dataset(s) and code	Monday, Nov 9	your GitHub project repo
Blog Post	Tuesday, Dec. 1 by 5pm	Website (using GitHub Pages)
1-on-1 presentation + Q&A	Week of Monday, Nov. 30	My office
Peer share	Tuesday, Dec. 8	In class
Reflection	Wednesday, Dec. 9 by 5pm	Gradescope

Getting Started

Review the project description on your own and review the American Association of Colleges and Universities' (AACU) Teamwork Value Rubric.

First Meeting

Meet with your team members as soon as possible to get some logistics out of the way and to start generating ideas. By the end of your first meeting, you should have done the following, in some order:

- Exchange contact information and decide on how you would like to communicate with each other.
- Discuss initial ideas (what were your top rankings? do you think you'll stick with that or are there other common interests worth exploring?)
- Come up with a team name (bonus points for a team image/logo)
- Set up project repo (instructions below)
- Set up a plan to complete your project proposal

Project Repo Setup

Within our course's GitHub organization, one team member should follow these steps to create a **New repository** from the provided template:

1. In the **Repository template** section, use the drop-down menu to select **acstat231-f26/blog-template**.
2. **Name** the repo using the convention *blog-team-name*, e.g. *blog-mighty-mammoths*
3. List your full names in the repo **Description** to help me keep track of things, e.g. "Brittney Bailey, Kara Yacoubou Djima, and Kat Correia"
4. Make the repo **Public**
5. Add a **.gitignore** template for **R**
6. Hit **Create repository**
7. Go to the repo **Settings** and add team members (only) as collaborators with **Admin** responsibilities. (I have access by default so you do not need to add me.)
8. Still in the repo **Settings**, click on **Pages** in the menu on the left-hand side of the page.
9. Under the **Branch** section where it currently says "None", click on the dropdown menu there to select **main**.
10. In the second dropdown that currently says "/(root)", click on the dropdown menu there to choose **/docs**. Then click **Save**. In a couple minutes, you will have a published website!
11. When the link pops up, copy the link and return to the main landing page of your repo to edit the **About** section.

12. Make sure the names of your team members are in the **Description** section and add the link to the website in the **Website** section.

All members should then take the usual steps to open the shared repo in GitHub Desktop and clone the repo to their STAT 231 course folder (where all your other course repos are).

Blog website starter

1. Open `index.qmd` and add your names to the author line. Save your changes and render the file to html.
2. In GitHub Desktop, **commit** both the `index.qmd` and the `index.html` files and **push** your changes back to github.com.
3. Visit your blog website link (e.g., https://acstat231-f26.github.io/blog-mighty-mammoths) to see that the author information (bottom left of page) updated correctly

Note: Do **NOT** change the name of `index.qmd` or `index.html`! These files must keep these names for your blog website to work.

Components

Initial proposal

The project proposal should be submitted as a new issue to GitHub (in your group project repo!) with "Proposal" in the title.

A proposal is not a vague, wishy-washy suggestion of what might possibly be done with the project. Rather, a proposal is a well thought-out plan for the project. It takes time to come up with ideas, test them out in a small way to see if they are feasible (are data available? if scraping, are you allowed to scrape? is it possible to scrape?), and revise or redirect as needed.

Feel free to talk to me and the SDS fellows if you have more than one direction in mind and would like feedback on different ideas before submitting the initial proposal. Part of the process here is learning how to refine questions, and how to evaluate whether you'll actually be able to answer the question you set out to answer.

As with your previous calendar project, your plans may change as your understanding of the data takes shape or as you run into issues and deadlines. That's okay! BUT we want to start with a good plan that seems feasible given available information at the start.

The issue body should contain the following content:

1. Working Title: Provide a working title for your project

2. Purpose: Describe the general topic/phenomena you want to explore, as well as some questions that you hope to address.

3. Data source and format: Identify one or more data sources that could be used in the project. What form is the data in (downloadable csv file? needs to be scraped from web?)? What do you imagine will be challenging about ingesting and/or wrangling the data?

4. Blog plan: Identify which of the methods from the second half of the semester you plan to incorporate (spatial data, text analysis, network science, and/or unsupervised learning) and how. Discuss any other graphs, tables, or statistical content you hope to incorporate. Will any of your visualizations or tables be dynamic? Provide as much detail as possible.

Revised proposal

The revised proposal should be submitted as a comment in response to my comments on the original GitHub issue "Proposal". Respond to the feedback received in the initial proposal.

Wrangled dataset(s) and code

At this stage, your data should be in the format needed to create the visualizations and summaries you'll present in the blog. You may have more than one wrangled dataset. For instance, if there are some outputs in your blog that require the data to be in long format and other outputs in your blog that require the data to be in wide format, then you'll want to create both a long format dataset (e.g., *healthcare_long.Rds*) and a wide format dataset (e.g., *healthcare_wide.Rds*) in your data wrangling script.

How do you know if you've wrangled enough? You should have wrangled your data to the point where you have minimal to no wrangling code within your blog code. If you find you need to add wrangling within the *index.qmd* file, **don't**—go back to your wrangling file, make updates there, and resave your data.

The (reproducible!) code to create the dataset(s) used in your blog should be saved in an R or qmd file within your group repository and appropriately named, e.g. *scrape.R* or *wrangle.qmd*. Include a note at the start of each file to summarize the content and describe each person's contribution to the wrangling process in that file.

Your wrangled dataset(s) that are ready for analysis within your blog should be saved in a *data* folder (in a separate location from the pre-wrangled data files) and loaded at the top of your `index.qmd` program.

If multiple datasets are scraped or wrangled independently by different people, it's okay to have separate scripts as long as the filenames clearly indicate the contents (e.g., you might have *scrape-wiki.R* and *scrape-weather.R*). You might also, after the fact, combine separate files into one larger file and delete the separate files.

Make sure your repository is organized and be sure your code is reproducible, readable, organized logically, and documented with informative comments. Where individual work is done, authorship should be indicated (at the top of the document and, if relevant, preceding the corresponding code sections).

For this checkpoint, be sure to push all your final R scripts and/or qmd files used to ingest and wrangle your code, as well as the final saved dataset(s) to your blog project repo on GitHub.

Your GitHub repo files should be up to date and organized. An example directory structure is:

```
shiny-mighty-mammoths                  (repo folder)
|
|--- data                              (subfolder)
|    |--- weather_sales.Rds
|
|--- docs                              (subfolder)
|    |--- index.html
|
|--- index.qmd
|
|--- wrangling                          (subfolder)
|    |--- data-scrape-weather.R
|    |--- data-wrangle-weather.qmd
|    |--- data-wrangle-sales.qmd
|    |--- data-combine-final.qmd
|
|--- raw-data                           (subfolder)
|    |--- my-raw-messy-sales-data.csv
```

Blog post

References

All data sources, any key R packages, and any other sources used in developing your blog (including AI) should be cited in full with relevant links in a list of references at the end of your blog. Your blog post should also link to these sources as they are discussed as either direct links or in-text citations.

For examples of data citations, see this resource on Citing data sources from Columbia University. Many data repository websites will provide details on how to cite specific datasets, but otherwise you should follow the guidance at the link.

Write-up

You do not need to follow a specific format in the blog post, but you should start with an introduction and finish with a conclusion and list of references. With the exception of the list of references, these components do not need to follow the formal writing style that you would use in most other classes. Here, a colloquial style that is accessible to a lay reader is appropriate.

Your write-up should address the following questions in some way:

- ☐ **What is this about?** Do not assume your readers have any specific knowledge of your subject matter! For example, if your project involves phylogenetic trees, you should assume your audience has only a very simple understanding of genetics.
- ☐ **Why should anyone care about this?**
- ☐ **What are the data?** What was the source of the data (who collected it, when, in what way, and why)? What kind of data was it? Is there a link to the data or some other way for the reader to follow up on your work? All data sources should be cited (include a list of references at the end of your blog).
- ☐ **What are your methods and findings?** What kind of statistical analysis did you do? What are your results/conclusions? Again, even if you display code showing how some of the analyses were performed, it is up to you to interpret, in simple terms, the results of these calculations (avoid statistical jargon to the degree possible when writing conclusions). Do not forget about units, axis labels, etc.

  > **BAD methods/interpretation example (too much jargon!!)**: We fit the regression model $\widehat{\text{Price}} = 120000 + 8(\text{square footage}) + 32152(\text{bedrooms})) - 5(\text{days on market})$ tested the null hypothesis $H_0 : \beta_2 = 0$ against the alternative hypothesis $H_a : \beta_2 \neq 0$ and our test statistics was $t = 2.47$ and our p-value, which is the probability of getting data as extreme or more extreme than the data we observed under the assumption the null hypothesis is true, was only 0.03 which is less than our $\alpha = 0.05$ signifiance level. So we reject the null hypotheis of $H_0 : \beta_2 = 0$ and conclude we have evidence for the alternative hypothesis $H_a : \beta_2 \neq 0$. And we conclude that `bedrooms` is a signficant predictor of `price`. $\hat{\beta}_2 = 31152$ mean for each additional bedroom the home price goes up \$32,152.
  >
  > **GOOD methods/interpretation example**: We used linear regression to model the sale price of homes in Amherst, MA based on the square footage, number of bedrooms of the home, and days on the market (see results in Table 2 [*Table 2 in this case would be a nicely formatted table of coefficients, standard errors and/or CIs, and p-values*]). We found that, after adjusting for square footage of the home and days on the market, for each additional bedroom a house has, the expected sales price increases by \$32,152 ($p = 0.03$). Additionally, after adjusting for number of bedrooms and days on the market, the expected sale price increases by \$8 per square foot ($p < 0.001$). The number of days the house has been on the market does not seem to have a significant effect on the sale price ($\hat{\beta} = -5$, $p = 0.53$).

☐ **What are the limitations of your work?** Be clear so that others do not misinterpret your findings. To what population do your results apply? Do they generalize? Could your work be extended with more data or computational power or time to analyze? How could your study be improved? Suggesting plausible extensions doesn't weaken your work; it strengthens it by connecting it to future work.

Code

The website's qmd file(s) should contain the minimal set of code necessary for your published website, and any extensive wrangling should be in separate R scripts.

The code does **NOT** need to be shown on the published web page. The default code chunk option will hide your code (`echo: false`). Only set `echo: true` for code you wish to show the audience. For example, if you used some nifty, new functions and/or some old functions in a creative way, you may want to showcase some of the code on the post as a way to teach the audience about these functions and techniques. However, the audience does not need to see every `filter()`, `mutate()`, `summarize()` etc. you use (unless the wrangling is part of the exposition!).

The code is there to support the technical reader who wishes to dig into your work, not to substitute for written explanation. Do not present long unbroken chunks of code without offering written explanations.

Be sure that the process is reproducible from a clean environment, with relative filepaths, and that all files needed to reproduce your work are available.

Visualizations and tables

Visualizations, particularly interactive ones, will be well-received. That said, do not over-use visualizations. You may be better off with one complicated but well-crafted visualization as opposed to many quick-and-dirty plots. Any plots should be well-thought-out, properly labeled, informative, and visually appealing.

If you want to include dynamic visualizations or tables, you should explore your options from packages that are built from **htmlwidgets**. These **htmlwidgets**-based packages offer ways to build lighter-weight, dynamic visualizations or tables that don't require an R server to run! A more complete list of packages is available on the linked website, but a short list includes:

- **DT**: Tabular data via DataTables
- **plotly**: Interactive graphics with D3
- **leaflet**: Interactive maps with OpenStreetMap
- **dygraphs**: Interactive time series visualization
- **visNetwork**: Network graph visualization vis.js
- **sparkline**: Small inline charts
- **threejs**: Interactive 3D graphics

More on Quarto website features

There are many many ways to customize your Quarto website. Many of the options are listed under the **Documents –> HTML** section of the Quarto Guide, but here are the main ones you might use frequently:

- HTML Basics for Quarto
- Changing a default theme

- How to link between pages within the website
- Customizing website navigation in the _quarto.yml file
- Creating an "About" page
- Using R HTML widgets in Quarto
- Bootstrap icons for Navigation bar

1-on-1 presentation + Q&A

You will sign up for a 20-minute block of time to meet with me in my office. I will have your blog pulled up on a shared screen. We'll spend the first 5 minutes with you introducing your project: your research question(s), data collection process, results, and conclusions. Then, we'll transition to a Q&A conversation where I'll ask you about your project, your code, your choices, and your findings.

There are several reasons why I've chosen this format for the major assessment in this course:

- **It emphasizes understanding rather than memorization or code production**. Data scientists routinely consult documentation, online resources, colleagues, and increasingly AI tools while they work. I expect you'll do the same while completing your project. What's most important is not whether every line of code came entirely from memory, but that you understand what your code is doing, why you chose your approach, and how to interpret your results. The conversation gives you an opportunity to demonstrate that understanding.

- **It develops scientific communication skills**. Data science is highly collaborative. You'll regularly need to explain analyses, justify decisions, answer questions, and communicate results to people with different backgrounds. This assessment gives you practice with those skills in a supportive setting. To help you prepare, we'll have a guest expert on public speaking, opportunities to practice in class (you'll even get to quiz me!), and time to receive feedback before your presentation.

- **It allows for richer feedback**. Unlike a written report alone, a conversation lets me ask follow-up questions, better understand your thinking, and give individualized feedback on both your statistical reasoning and communication. If something in your report is unclear, you'll have a chance to clarify it rather than losing credit because I guessed incorrectly.

- **It helps us build relationships**. Data science often involves long hours of coding on a computer, but the field is fundamentally collaborative. The best ideas emerge through conversation, asking questions, explaining your thinking, and learning from one another. As AI makes it easier to write code and perform technical tasks, these distinctly human skills – communicating clearly, building trust, collaborating effectively, and connecting with other people – become even more valuable. I also don't want our class to be defined solely by interactions with a screen. My hope is that this assessment encourages you to come to office hours, visit the SDS Fellows office hours, practice with your classmates, and experience the joy of discussing ideas face-to-face. Those relationships often become one of the most rewarding parts of learning, and they're a skill you'll carry well beyond this course.

Peer Share

The Peer Share is an opportunity to learn from the wide range of questions, data sources, methods, and approaches taken by your classmates. You will thoughtfully explore two other groups' blogs and participate in small-group conversations about their work. You'll be asked to share what you found interesting or

effective, ask questions, and identify ideas that you might carry forward into your own future data science work. Prompts will be provided.

Reflection

The reflection will be completed individually, and consists of a series of questions designed to help you reflect on the trajectory of your group's work together.